# Project Build Report

What has been built, what has been verified, and the exact state of the machine-learning component

Repository: juntinggaa/hedge-fund-positioning (private) | Branch: main | Head commit: ad60892
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## 1. Executive summary

Read this section first; it contains the one thing that matters most.

**The machine-learning model is specified, scaffolded, tested for leakage safety, and not fitted.** No estimator has been trained on this data. There is no out-of-sample $R^2$, no residual series, no positioning "surprise" and no predictive result, because the fourteen functions that would produce them raise `NotImplementedError` by design. What exists is the statistical architecture around the model: the walk-forward splitter that decides which observations a model may ever see, the as-of-date and release-date logic that defines a legitimate prediction window, and the content-hash machinery that pins the data vintage a fit would be reproducible against. That architecture is complete and covered by 58 passing tests.

This is a deliberate ordering, not an abandoned effort. The leakage guards were built first because a fitted model with a leak is worse than no model: it produces a number that looks like a finding.

**The role of the model also changed mid-project, and this is the most important design fact in the repository.** The original project made the ML model the result: decompose leveraged-fund Treasury positioning into an explained component and a residual, then test whether the residual predicts returns. Under the course requirements the project was re-scoped to an audit of how large language models interpret institutional positioning data. The ML model survived the re-scope but changed job entirely. It is no longer asked to be *right*. It is now a **stimulus generator**: its expected-position and residual output is the experimental material handed to an LLM in two of four conditions, precisely to find out whether a plausible-looking quantitative output causes the model to manufacture a causal story the number cannot support.

That inverts the usual quality requirement. A forecasting project needs the model to be accurate. This project needs it to be *honest about what it is*, because the thing under test is whether an LLM respects the limits of a statistic that asserts nothing about causation.

**What is verified and finished:** the data layer. 1,057 weekly CFTC observations for 10-Year Treasury Note futures, 2006-06-13 to 2026-09-08, reconciled against CFTC's own official archives with zero mismatches, pinned by SHA-256 content hash which still matched on the report date.

**What is evidence, not plan:** three pilot LLM runs were executed and the project's original central hypothesis was falsified by them. That falsification is reported in the proposal rather than hidden, and the research design was rebuilt around what the pilots actually showed.

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## 2. Module-by-module state

```
File                              Lines  State         Tests
-------------------------------------------------------------------
src/hfpos/ingest/cftc.py            254  implemented   yes
src/hfpos/ingest/cftc_archive.py    126  implemented   yes
src/hfpos/manifest.py                90   implemented   yes
src/hfpos/timing.py                  85   implemented   yes
src/hfpos/stage1/walkforward.py      80   implemented   yes
src/hfpos/config.py                  28   implemented   indirect
scripts/draw_sample.py               89   implemented   no
scripts/run_pilots.py               125  implemented   no
-------------------------------------------------------------------
src/hfpos/features/build.py          15   scaffold      no
src/hfpos/features/basis.py          18   scaffold      no
src/hfpos/features/momentum.py       15   scaffold      no
src/hfpos/features/macro.py          15   scaffold      no
src/hfpos/stage0/leadlag.py          15   scaffold      no
src/hfpos/stage1/models.py           17   scaffold      no
src/hfpos/stage1/run.py              19   scaffold      no
src/hfpos/stage2/predict.py          24   scaffold      no
src/hfpos/stage2/inference.py        19   scaffold      no
src/hfpos/eval/metrics.py            15   scaffold      no
src/hfpos/eval/bootstrap.py          16   scaffold      no
src/hfpos/eval/plots.py              17   scaffold      no
src/hfpos/ingest/prices.py           16   scaffold      no
src/hfpos/ingest/fred.py             16   scaffold      no
```

"Scaffold" means: module docstring stating the method, the known traps and the decision rationale, plus a typed function signature that raises `NotImplementedError` with its issue number. Fourteen such call sites exist. They are deliberately loud: nothing silently returns a placeholder number that could leak into a write-up.

**Test status, re-run on the report date:** 53 offline tests pass, 5 network tests pass, 58 total, no failures, no skips. The network tests are the ones that hit the live CFTC endpoint and re-check the reconciliation, so passing them means the cross-source agreement still holds today and is not a stale claim from September.

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## 3. The machine-learning model

### 3.1 What it is asked to do

One supervised model, one target, one job: predict what leveraged-fund positioning in 10-Year Treasury futures *would ordinarily be*, given observable market conditions in that week. The part it cannot predict is the residual.

The model is explicitly **not** a strategy decomposition. It does not and cannot separate basis trades from directional bets, because the cash leg of the basis trade is not in the CFTC futures dataset at all. This distinction is the entire reason the project exists in its current form, and it is stated three ways in the proposal: explained is not caused; total model fit is not the contribution of any single regressor; and the residual is not directional, since it also contains misspecification, omitted variables and data error.

### 3.2 Specification

```
Target        d(Net / Open Interest), leveraged funds, weekly first difference
Horizon       contemporaneous (the model explains this week's change)
Estimators    Ridge, Lasso, XGBoost -- all three reported, not just the winner
Features      lagged returns 1/4/12w; realized volatility; yield levels and
              changes (2/5/10/30Y); 10Y-2Y slope; curvature; SOFR-IORB spread;
              cash-futures basis proxy; change in open interest; lagged own
              position level and change; VIX; roll-week dummy; FOMC-week dummy
Validation    expanding window, min 260 weeks of training, refit every 13 weeks
Config        all of the above lives in config.yaml, not in code
```

**Why the target is scaled by open interest rather than raw contracts.** Open interest in this contract is not stationary across the sample: it ranges from 888,595 contracts (week ending 2009-02-24) to 6,254,472 (2026-05-26), a factor of **7.04**. A model trained on raw net contracts would spend its capacity learning that the market got bigger. Note one correction to the earlier project plan, which described this growth as "an order of magnitude": the verified figure is 7.0x, not 10x. The design decision is unaffected; the wording was loose.

**Why the first difference rather than the level.** The level of net/OI has a lag-1 autocorrelation of **0.991**. Predicting it is almost free and almost meaningless: last week's value is a near-perfect forecast. The first difference has a lag-1 autocorrelation of **-0.154**, which is a genuine prediction problem. This is also why the LLM sample of 40 weeks is spaced at least eight weeks apart -- at 0.991, adjacent weeks are near-duplicate stimuli.

**Three estimators, all reported.** With roughly 1,000 observations and 20-40 features, the linear models may well beat XGBoost. The plan commits in advance to reporting that if it happens, as a finding about the data rather than a failure of the project. Nothing in the write-up depends on the gradient-boosted model winning.

### 3.3 The walk-forward estimator: the one ML component that is finished

`src/hfpos/stage1/walkforward.py` is implemented and is the single point through which every guard against look-ahead bias routes. Two properties are enforced by tests.

**Property 1 -- training data lies strictly before the test block.** No fold ever trains on an observation at or after the first predicted week.

**Property 2 -- an embargo gap separates them.** This is the subtle one. A forward-looking target labelled at week *t* is not fully observed until week *t+h*. Training on weeks within *h* of the test block therefore leaks the answer, even though those weeks are nominally in the past. Stage 1 targets are contemporaneous and take embargo 0; any Stage 2 test at horizon *h* must pass embargo *h-1*.

```
def split(self, n):
    idx = np.arange(n)
    for start in range(self._first_test_start(), n, self.step):
        train_end = start - self.embargo
        yield Split(train=idx[:train_end], test=idx[start : start + self.step])
```

Verified behaviour on the real sample length of 1,057 weeks: **62 folds**, **797 predicted weeks** at embargo 0, and **794** at embargo 3 (the 4-week-horizon setting). Test blocks tile the tail of the sample contiguously -- no gaps, no overlap, every eligible observation predicted exactly once -- which is what makes the out-of-sample $R^2$ a single honest number rather than an average over arbitrary splits.

The accompanying rule, stated in the module and in the plan: anything fitted on training data -- feature scalers, hyperparameters, feature selection -- is fitted **inside** the loop, per fold. There are no exceptions, and this is the most common way published forecasting results turn out to be wrong.

### 3.4 The two gates the model must pass before it is allowed into the experiment

These are pre-committed, and either one can kill the ML component.

**Gate 1 -- Stage-1 out-of-sample $R^2$.** If it is near zero, the model explains nothing, the residual is just the raw series with extra steps, and there is no decomposition to hand the LLM.

**Gate 2 -- correlation between the residual and the raw change in net/OI.** If this exceeds roughly 0.9, the "removal of normal positioning" is cosmetic. The plan requires this number in the main text, not an appendix, on the reasoning that any competent reader will compute it themselves.

Both gates are scheduled before the arms that depend on them, and the proposal already names the fallback if they fail: ship the two arms that need no model (raw data, and qualitative market-structure

context), which still constitute a baseline, a controlled variation and two testable hypotheses.

### 3.5 Statistical hazards the design handles

**Generated-regressor problem.** In the original Stage 2, the regressor *is itself an estimate*, so ordinary standard errors are wrong (Pagan 1984). With an ML first stage there is no analytic correction. The design's answer is to block-bootstrap the **whole pipeline**, refitting Stage 1 inside every replication: stationary bootstrap, mean block 13 weeks, 1,000 replications, seed 20260918. Bootstrapping only the second stage does not fix this and is the most likely point of attack in an oral examination.

**Overlapping returns.** The 2- and 4-week horizons overlap, so residuals are serially correlated by construction. Newey-West with lag >= h-1, or Hodrick standard errors.

**Multiplicity.** One pre-registered primary test, unadjusted. Everything else -- 3 groups x 4 horizons x 4 maturities is roughly 48 tests -- is adjusted (Romano-Wolf or Benjamini-Hochberg) with the family size stated. At 5% you expect two or three false positives in that family by construction.

**Power, stated in advance.** Roughly 797 test weeks, but only about 199 non-overlapping ones at a 4-week horizon. A realistic incremental $R^2$ in this literature is 1-2%. The plan writes that down beforehand so that a small effect is read as a small effect rather than as failure.

### 3.6 The failure mode the model is now designed to provoke

In the re-scoped project the model's output becomes an experimental stimulus, and the specific error being hunted is **causalisation of a residual**. Given an actual position of -3.2m, an expected -2.6m and a residual of -0.6m, an LLM may report something like *"81% is attributable to basis trading, 19% to directional bearishness."*

Every element of that sentence is unwarranted, and the model asserted none of it. The chain the project traces is:

> *quantitative tool -> apparently precise output -> LLM causalises the statistic -> confident explanation -> user sees unjustified certainty*

Two of the four experimental conditions exist to measure this: one hands over the expected-position and residual numbers unlabelled, the other hands over the identical numbers plus one sentence stating what a residual does and does not mean. If the single sentence repairs the failure, the problem is a design choice rather than an inherent property of fluent generation -- which is a sharper and more useful result than a general complaint about hallucination.

### 3.7 What finishing the model actually requires

In dependency order, with the hard part named honestly.

| | | | |
|---|---|---|---|
| 1 | Price ingest and roll handling | prices.py | ZN=F is a stitched front-month series; naive weekly returns inject a false jump at every quarterly roll |
| 2 | FRED ingest | fred.py | routine |
| 3 | Basis proxy | basis.py | THE HARD ONE -- see below |
| 4 | Momentum and macro features | momentum/macro | routine |
| 5 | Feature assembly onto the Tuesday grid | build.py | alignment care needed |
| 6 | Lead-lag check (do positions follow returns?) | leadlag.py | must run BEFORE any model is fitted |
| 7 | Estimators and walk-forward driver | models/run.py | harness already exists |
| 8 | Gate diagnostics | run.py | decides the rest |

**Item 3 is harder than everything else combined.** The claim "we modelled conditions favouring basis trading" rests entirely on a proxy for the cash-futures basis, and the preferred construction (cheapest-to-deliver implied repo against general-collateral repo) needs the deliverable basket and

conversion factors. The fallback ladder, best first, is: CTD implied repo vs GC repo; futures-implied yield vs cash yield; swap spreads; SOFR-IORB spread. SOFR data begins only in 2018, so the early sample needs a lower rung regardless. And the basis trade was small before roughly 2015, which means "conditions favouring basis trading" is not the same quantity in 2008 as in 2024. The repository records which rung was used and requires the claim to be downgraded to match.

**Item 6 is a trap worth naming.** If leveraged-fund positioning merely *follows* returns, then a feature set without lagged returns produces a "surprise" that is really just momentum in disguise. That is why the lead-lag check is sequenced before any fitting rather than reported as a robustness check afterwards.

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## 4. The data layer

This part is finished, and it is the part the rest of the project stands on.

```
Source        CFTC Traders in Financial Futures, futures-only (dataset gpe5-46if)
Contract      10-Year Treasury Note futures, market code 043602  [flagship]
Also pinned   2Y 042601 | 5Y 044601 | 30Y 020601
Coverage      1,057 weekly observations, 2006-06-13 to 2026-09-08
Integrity     SHA-256 content hash, re-verified against the manifest on 2026-11-03: MATCH
Cross-check   reconciled against CFTC's official annual archives across 2,310 cells
              in 2012, 2019, 2024 and 2025 -- zero mismatches
```

**The contract-code trap, found the hard way.** CFTC renamed the 10-Year contract in February 2022, from "10-YEAR U.S. TREASURY NOTES" to "UST 10Y NOTE". Filtering the dataset by contract *name* returns 817 rows or 240 rows instead of 1,057 -- silently, with no error. The ingest therefore filters by numeric market code only, and a test asserts that a placeholder code fails loudly rather than returning an empty frame.

**Two data irregularities, recorded rather than smoothed over.** 14 of the 1,057 snapshots are not dated on a Tuesday (holiday weeks shift the as-of date), and 22 week-to-week gaps are irregular. An earlier version of the timing module raised an exception on any non-Tuesday as-of date, which means it would have crashed on live data; it now resolves the release date as the next Friday from any business day. Both counts are stored in the data manifest.

### Verified descriptive facts

These are the numbers the LLM experiment's stimuli are drawn from, recomputed on the report date.

```
Leveraged funds net SHORT                  695 of 1,057 weeks   (65.8%)
  mean net position when short                        -17.0% of open interest
Leveraged funds net LONG                   362 of 1,057 weeks
Asset managers net LONG                           75.1% of weeks
  mean net position when long                    +18.2% of open interest
Most negative week in the sample           2025-09-02, -48.53% of open interest
Lag-1 autocorrelation of net/OI                                  0.991
Lag-1 autocorrelation of d(net/OI)                              -0.154
Open interest range                 888,595 to 6,254,472 contracts (7.04x)
```

The 362 net-long weeks matter for the experiment design: they supply the **sign-flip control**. If an LLM produces the basis-trade narrative on a week when leveraged funds were net long, the narrative is template-driven rather than evidence-driven.

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## 5. The Milestone 1 research design, in brief

**Research question.** When LLMs interpret institutional Treasury futures positioning, how reliably do they distinguish observed evidence, statistical inference, and claims about trader motive -- and does

quantitative model output improve that distinction or create new forms of false causal confidence?

**Four conditions**, run at a frontier tier and a free tier:

```
A        CFTC positioning and observable market data only
B        A + market-structure mechanism, stated WITHOUT any magnitudes
C-bare   A + ML expected position and residual, unlabelled
C-ann    C-bare + one sentence on what a residual does and does not mean
```

Condition B must state the mechanism without asserting a proportion. If it hands over a magnitude, it has given away the very claim the experiment measures and the comparison with the C conditions is confounded.

**Primary outcome is which boundary was crossed, not a single error rate.** Three boundaries are coded separately -- data to statistical inference, inference to mechanism, mechanism to magnitude -- each at severity 0 to 3, with the scale anchored on verbatim pilot responses so coders calibrate against evidence rather than intuition. A model can clear the first two boundaries and fail the third, which is exactly what the pilots showed; one combined rate would have hidden the finding.

**Sampling and scale.** 40 weeks, ten per stratum, stratified across position state and basis-trade era, spaced at least eight weeks apart, drawn by a seeded script that writes the full list to `docs/milestone1/sample.csv`. 40 weeks x 4 arms x 2 tiers = **320 responses**. The pilot week is held in a separate purposive frame rather than quietly counted in the stratified draw, because it sits 7 days from a drawn week and would violate the spacing rule.

**Analysis.** Paired design, so the primary test is McNemar on discordant pairs per contrast. Power depends on how many cases change between arms rather than on n. Absolute rates carry roughly a +/-16 percentage-point margin at n=40: adequate for large effects, not for subtle ones. The proposal says so explicitly and calls this an exploratory audit rather than a population estimate.

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## 6. Pilot evidence, and the hypothesis it killed

Three pilot runs were executed, all on the week ending 2 September 2025, the most extreme week in the entire sample.

```
#  Arm         Tier       Run by          Result
-------------------------------------------------------------------------
1  A bare      frontier   me, via browser hedged, severity 1
2  B grounded  frontier   me, via browser hedged
3  A bare      free       the user        OVERCLAIMED, severity 2
```

**The frontier model, with no prompting about basis trades, said:** that hedge funds often short Treasury futures as one leg of the cash-futures basis trade, that such a position is not necessarily a bet that yields will rise, and that it would therefore not conclude from the CFTC numbers alone that hedge funds expect yields to soar. It also computed -48.5% of open interest exactly.

**The free model, on identical data, asserted** that the bulk of the position reflects execution of the Treasury basis trade -- a proportional claim its input cannot support, since the cash leg is absent from the data -- and described open interest as "over 60%" where the verified figure is 59.88%.

**Consequence: the original hypothesis was falsified before any of the seven project weeks were spent on it.** The naive prediction was that models would misread a large short as a directional bearish view. They did not. The real failure is narrower and more interesting: **boundary-crossing** -- stating inference in the register of observation. The research design was rebuilt around that, and the falsification is reported in the proposal rather than quietly dropped.

The pilot's own limits are recorded with it: n=1 per cell, a web interface that exposes no model version string or token counts, no configurable sampling parameters, and "Unpersonalized" mode, which is not the mode a real retail user is in.

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## 7. Verification log

What was actually checked, and by what means.

```
Claim                                 How verified                     Status
-------------------------------------------------------------------------------
58 automated tests pass               re-run on report date            CONFIRMED
Data vintage unchanged                SHA-256 vs pinned manifest       CONFIRMED
API matches CFTC official archives    2,310 cells, 4 years             CONFIRMED
Contract codes correct                live endpoint + failing-code test CONFIRMED
Descriptive statistics in this report recomputed from cached parquet   CONFIRMED
Citations (9 sources)                 publisher records, 18 Sep 2026    bibliographic
                                                                        details only
Pilot responses                       run live, logged verbatim        CONFIRMED
ML out-of-sample performance          NOT RUN -- no model fitted       OPEN
```

**Things this report deliberately does not claim.** No ML performance number exists, so none is quoted. The nine citations have had their bibliographic details verified at source and one retrieved in full, but their *substantive* claims still need checking against full text before submission. The free-tier run of condition B -- the missing fourth cell of the 2x2 -- has not been executed, and it speaks directly to one of the hypotheses.

Two corrections I made during the work are worth recording, since both were my errors: I overstated the roll-date risk in the price series (tested afterwards: roll months account for 20% of the largest weekly moves, below the 33% baseline you would expect by chance), and I claimed sampling temperature would be "fixed and logged" when current frontier models no longer accept a temperature parameter at all. The proposal now runs replicates and reports variation instead.

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## 8. Open items

```
1  Fill bracketed placeholders in the proposal: teammate names, date, owners
2  Run condition B at the free tier -- the missing cell of the 2x2
3  Trim the appendix from 2 pages to the 1-page limit
4  Upload the .docx to Google Drive for teammate editing (needs a human; no
   connector is attached to this session)
5  Verify substantive claims of all 9 citations against full text
6  Retire or rewrite the 18 GitHub issues that still describe the abandoned
   forecasting project
7  Confirm the re-scope with the teaching team if they are reachable
```

Items 1 and 2 are the only ones that block submission quality. Item 2 is the one that would most improve the evidence base, and it costs a single prompt.

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## Appendix: reproducing everything in this report

```
# install dependencies into the project venv
uv pip install -r requirements.txt pytest

# offline tests (53)
.venv/bin/python -m pytest -q

# live-endpoint reconciliation tests (5)
.venv/bin/python -m pytest -m network -q

# redraw the stratified 40-week sample (deterministic, seed 20260918)
.venv/bin/python scripts/draw_sample.py

# the LLM pilot harness (needs ANTHROPIC_API_KEY, not set in this environment)
.venv/bin/python scripts/run_pilots.py --replicates 3
```

Every number the analysis depends on lives in `config.yaml` rather than in code, and the data vintage is pinned by content hash in `data/manifests/`. A fit that cannot be reproduced against that hash is not a result.